

String Theory and M-Theory

Lecture 2: From Boundary Conditions to the Open-String Spectrum

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Chapter 1

From Boundary Conditions to the Open-String Spectrum

Overview. Our aim is to quantize an open string and decide whether its states deserve to be called particles. The route is elementary but exacting. We first prepare the calculus and Fourier analysis of functions on $0 \leq \sigma \leq \pi$. We then replace pointlikeness by a spectral criterion for particle behavior, review the infinite-momentum description in which the transverse dynamics become nonrelativistic, and build the string from a chain of light masses and stiff springs. The continuum string is a wave system with free endpoints. Its Neumann modes decouple into harmonic oscillators with frequencies $1, 2, 3, \dots$; quantization therefore produces a discrete, gapped spectrum. String interactions finally force us beyond open strings: an open string can join its own endpoints and become closed.

1.1 Calculus on the open-string interval

The calculation will repeatedly turn discrete sums into integrals and nearest-neighbor differences into derivatives. It is better to put those facts in place before they interrupt the physics.

Let $x(\sigma)$ be a function on

$$0 \leq \sigma \leq \pi. \quad (1.1)$$

The notation anticipates string theory: σ labels position along the string, while x will become one of its transverse coordinates. Divide the interval into N pieces,

$$\Delta\sigma = \frac{\pi}{N}, \quad \sigma_i = i \Delta\sigma, \quad x_i = x(\sigma_i). \quad (1.2)$$

For a smooth limiting function,

$$\Delta x_i \equiv x_i - x_{i-1} = \frac{\partial x}{\partial \sigma} \Delta\sigma + O((\Delta\sigma)^2), \quad (1.3)$$

and the Riemann sum becomes

$$\sum_i x_i \Delta\sigma \longrightarrow \int_0^\pi x(\sigma) d\sigma. \quad (1.4)$$

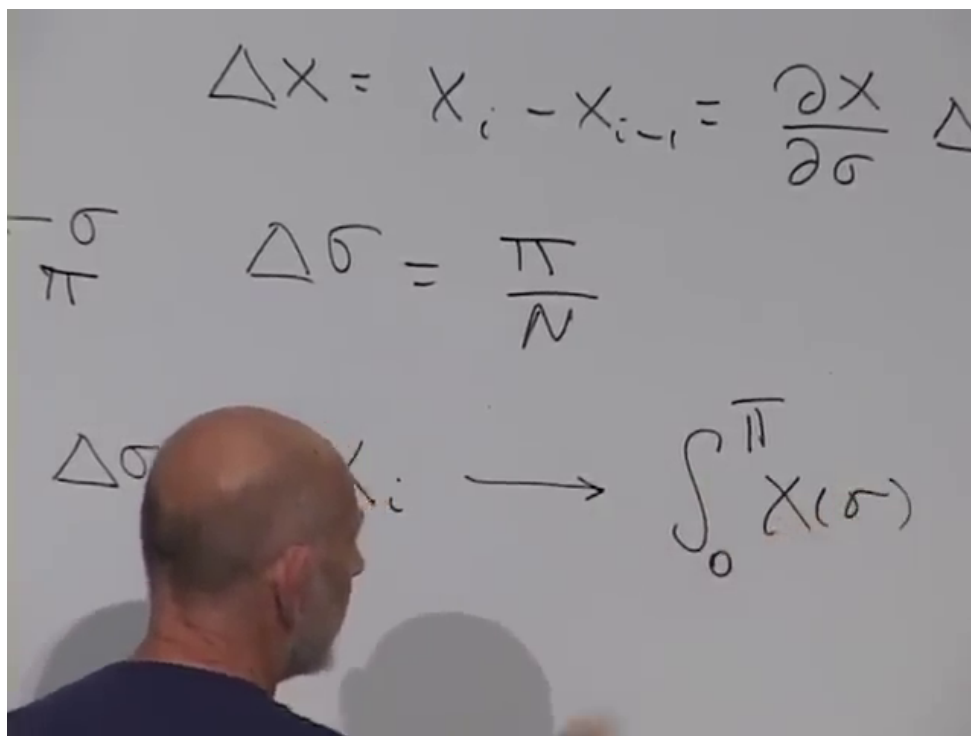


Figure 1.1: The finite-difference and Riemann-sum rules kept on the board for the later continuum limit.

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These are definitions familiar from calculus. Their importance here is in the scaling: Δx_i brings one power of $\Delta\sigma$, whereas a sum brings one inverse power when it becomes an integral.

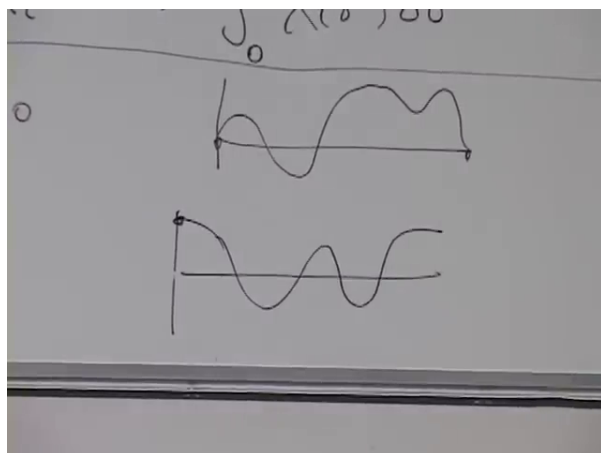
1.1.1 Dirichlet and Neumann endpoints

Fourier analysis on a finite interval begins with a choice at the endpoints. Two choices will recur.

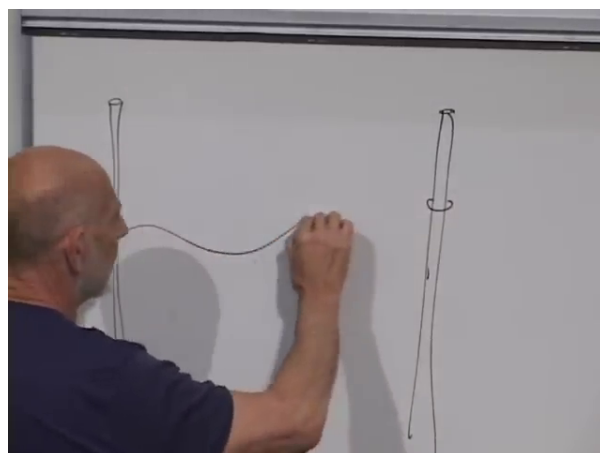
$$\text{Dirichlet:} \quad x(0) = x(\pi) = 0, \quad (1.5)$$

$$\text{Neumann:} \quad \partial_\sigma x|_{\sigma=0} = \partial_\sigma x|_{\sigma=\pi} = 0. \quad (1.6)$$

A violin string held at both ends gives the Dirichlet picture: its displacement vanishes at the boundaries. A Neumann function need not vanish there; it meets each boundary with zero slope. The two characteristic shapes are visible together on the board.



(a) Pinned and flat endpoint profiles.



(b) Light rings sliding on vertical poles.

Figure 1.2: Boundary conditions as shapes and as mechanics. The sliding-ring construction gives a free endpoint an unambiguous transverse motion.

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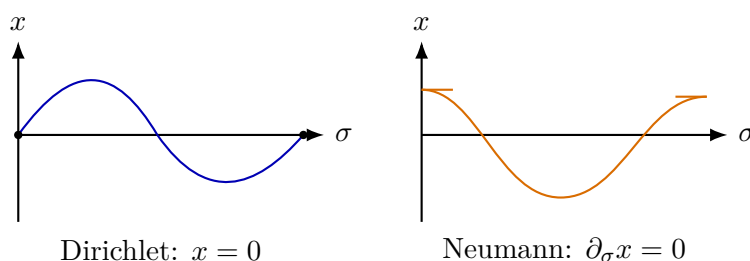


Figure 1.3: The value is fixed for Dirichlet data; the slope is fixed for Neumann data.

Question from the audience. Does an open or closed organ pipe supply a clean example of one of these boundary conditions?^a

Response. It can, but one must first say which acoustic variable is being expanded. Pressure perturbation and longitudinal displacement obey complementary idealized conditions at an open end. Rather than settle that side question at the board, use the sliding-ring model: nearly massless rings can move vertically on the poles, so the string endpoints are free and the Neumann picture is direct.

^aLecture timestamp: 00:09:19.

Question from the audience. Can the graph rise vertically and contain a sharp jump?^a

Response. Fourier analysis permits isolated jumps in a piecewise continuous function. A physical string is more restrictive. A jump disconnects neighboring points and makes the spatial derivative $\partial_\sigma x$ singular in the continuum energy derived below. We therefore restrict attention to profiles that are continuous and differentiable.

^aLecture timestamp: 00:09:53.

Question from the audience. Why should the open end of a pipe, or any free end, satisfy a boundary condition at all?^a

Response. The acoustic example depends on the chosen field and was left unresolved in the discussion. The mechanical example does not. A very light ring is free to slide but is still attached to the final string segment. We shall derive its condition from Newton's law rather than assume it: finite endpoint acceleration forces the slope to vanish.

^aLecture timestamp: 00:11:14.

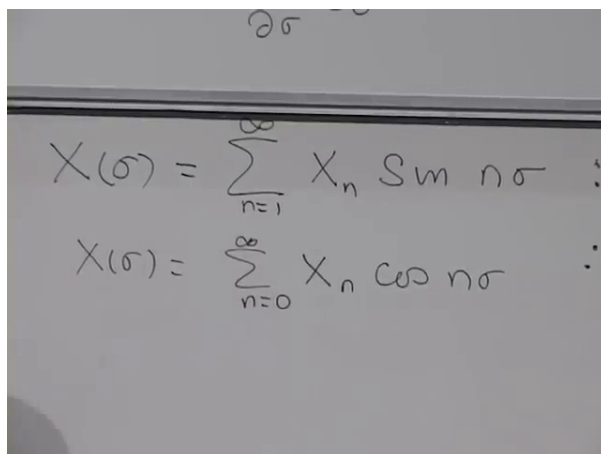
1.1.2 The Fourier bases

Once the endpoint class is fixed, the corresponding Fourier basis is fixed. On $0 \leq \sigma \leq \pi$,

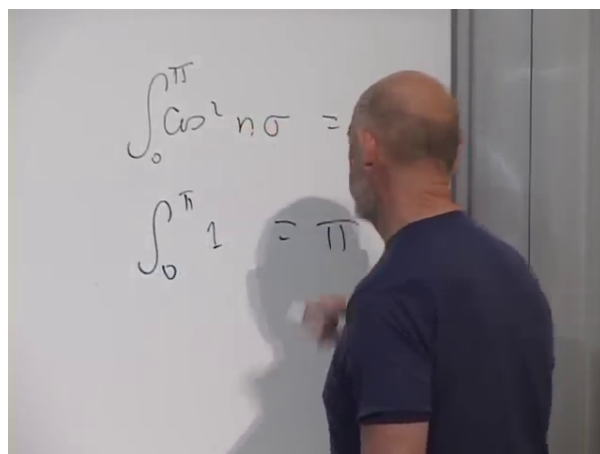
$$x_D(\sigma) = \sum_{n=1}^{\infty} x_n \sin(n\sigma), \quad (1.7)$$

$$x_N(\sigma) = \sum_{n=0}^{\infty} x_n \cos(n\sigma). \quad (1.8)$$

Every sine in (1.7) vanishes at both endpoints. Every cosine in (1.8) has zero endpoint derivative. The lower limits differ for a reason: $\sin(0\sigma) = 0$ contributes nothing, while $\cos(0\sigma) = 1$ is the constant, or zero, mode. It records the average value of a Neumann function.



(a) Sine and cosine expansions.



(b) Orthogonality and the exceptional constant mode.

Figure 1.4: The Fourier facts used later to diagonalize the string.

Lecture frame: 00:16:15 and 00:20:30.

Differentiation exchanges the two classes:

$$\begin{aligned} \partial_\sigma \sin(n\sigma) &= n \cos(n\sigma), \\ \partial_\sigma \cos(n\sigma) &= -n \sin(n\sigma). \end{aligned} \quad (1.9)$$

Thus the derivative of a Dirichlet series has Neumann behavior and the derivative of a Neumann series has Dirichlet behavior. Orthogonality then removes cross-terms between distinct modes:

$$\int_0^\pi \cos(n\sigma) \cos(m\sigma) d\sigma = \frac{\pi}{2} \delta_{nm}, \quad n, m \geq 1, \quad (1.10)$$

$$\int_0^\pi \sin(n\sigma) \sin(m\sigma) d\sigma = \frac{\pi}{2} \delta_{nm}, \quad n, m \geq 1, \quad (1.11)$$

$$\int_0^\pi 1 d\sigma = \pi. \quad (1.12)$$

The last line is the exceptional $n = m = 0$ cosine norm. A constant is not normalized like a nonzero harmonic, and that factor of two will distinguish the string's center-of-mass mode from its internal modes.

1.1.3 One oscillator formula to remember

After rescaling a coordinate so that its mechanical mass is one, a Hooke-law oscillator has

$$E_{\text{osc}} = \frac{1}{2} \dot{q}^2 + \frac{1}{2} \omega^2 q^2, \quad (1.13)$$

$$L_{\text{osc}} = \frac{1}{2} \dot{q}^2 - \frac{1}{2} \omega^2 q^2. \quad (1.14)$$

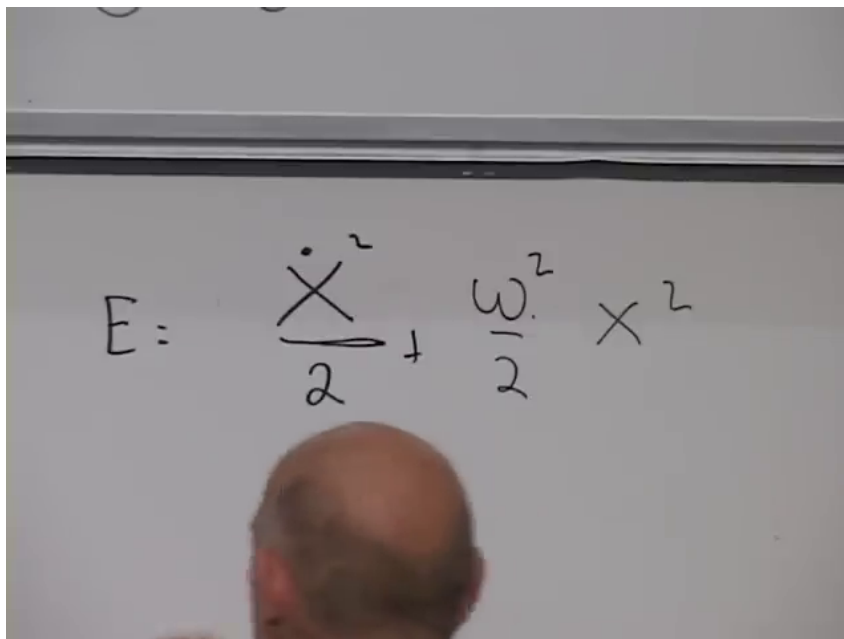


Figure 1.5: The oscillator energy, written so that the coefficient of q^2 immediately identifies ω^2 .

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Classically, $\omega = \sqrt{\kappa/m}$; the coordinate rescaling has set $m = 1$. Quantum mechanically,

$$E_Q = \hbar \omega \left(Q + \frac{1}{2} \right), \quad Q = 0, 1, 2, \dots \quad (1.15)$$

For level spacings we may subtract the common ground-state term and say that each quantum costs $\hbar\omega$. The string calculation will initially do exactly that and postpone the total zero-point energy.

Question from the audience. How is the oscillator frequency read from the energy formula?^a

Response. Before rescaling it is $\omega = \sqrt{\kappa/m}$. With unit mass the potential term is $\frac{1}{2}\omega^2 q^2$, so its quadratic coefficient is the square of the frequency.

^aLecture timestamp: 00:23:21.

1.2 What makes a composite system a particle?

A string is extended and will be modeled by many coupled pieces. In what sense can one of its quantum states nevertheless be a particle? Pointlikeness cannot be the decisive criterion. Protons and neutrons are composite. Electrons and photons are accompanied by quantum-field dressing even though no intrinsic substructure has been resolved. A classical geometric point surrounded by nothing is not the right image of a quantum excitation.

Nor do mass and center-of-mass position suffice. A sealed box of gas has both. A cup of coffee has both. We do not normally call either one a particle. Indivisibility also fails because a proton is divisible, and spin does not distinguish microscopic particles from macroscopic bodies.

The useful criterion is spectral isolation. Put the rest energy on a vertical axis and ask how much energy is required to reach another internal state. An electron has no experimentally accessible tower of electron excitations just above it. A proton does have resonances, but the first ones cost a few hundred MeV, a substantial fraction of the proton mass. Those states are separated enough to be individually identifiable.

A cup of coffee is the opposite. One can add an arbitrarily small amount of energy to a long-wavelength sound mode, nudge a molecule, or produce a phonon. There are enormously many states packed into a tiny interval. On laboratory scales they form a continuum. This is the practical distinction between a particle and *mush*.

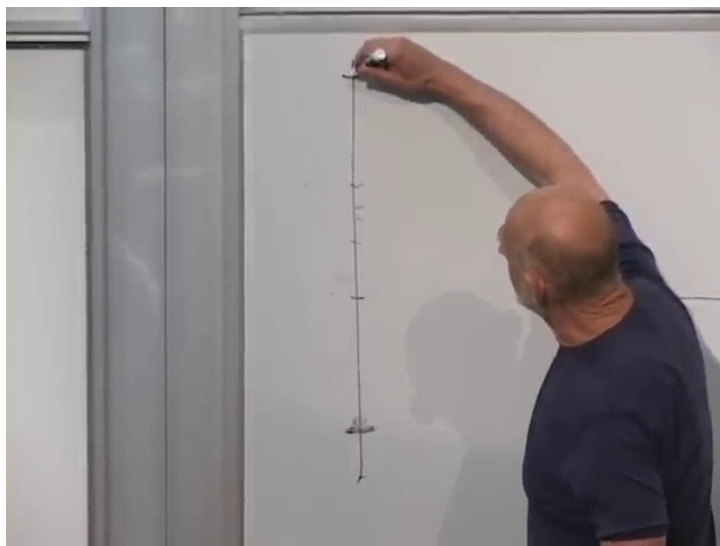


Figure 1.6: The board's vertical energy axis during the comparison of an isolated particle level with the dense excitations of ordinary matter.

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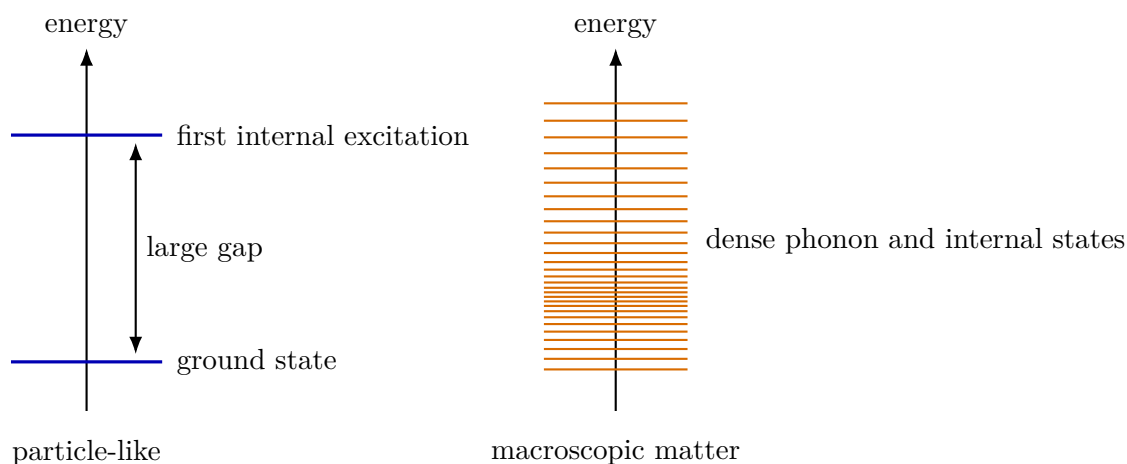


Figure 1.7: Spectral isolation, not literal pointlikeness, is the operational particle criterion used in the lecture.

This criterion turns the philosophical question into a calculation. Quantize the string, find its internal energy levels, and inspect their spacing. If the first excited string state lies near the Planck scale while the observed ground state is light, the gap exceeds present accelerator energies by more than twenty orders of magnitude. The string then behaves as an elementary particle throughout the accessible regime.

Question from the audience. Is particle behavior merely an issue of observational scale? If we zoom in far enough, do the levels cease to be separate?^a

Response. Zooming in may reveal that an object has many constituents, but it does not erase the spectrum. The relevant comparison is between the energy resolution of the experiment and the gap to the first internal excitation. A composite object can remain particle-like when that gap is enormous.

^aLecture timestamp: 00:33:33.

Question from the audience. Why is an ordinary violin string not a particle?^a

Response. One vibrational quantum carries an immeasurably small energy compared with the rest energy of the macroscopic string. Many quantum states therefore fit into a tiny relative interval. It belongs on the dense side of the spectral comparison, not the isolated side.

^aLecture timestamp: 00:34:37.

Question from the audience. Are photons and electrons point particles if quantum fields surround them with virtual fluctuations?^a

Response. The classical word *point* is doing too much work. The physical electron and photon are dressed field excitations, yet experiments have found no intrinsic constituent structure or finite form factor for either. What matters for the present argument is that neither has a dense tower of almost-degenerate internal states.^b

^aLecture timestamp: 00:35:40.

^bEditorial clarification: The lecture associates the electron's anomalous magnetic moment with a measurable quantum fuzz. In modern terminology the anomalous moment is a radiative dressing effect, not evidence for a resolved finite electron radius. Pointlike conventionally means that no intrinsic substructure or form factor has been detected down to the tested scale.

Question from the audience. Does the size of a photon depend on its frequency?^a

Response. Its wavelength certainly does. That is distinct from an intrinsic constituent size. Changing the frequency changes the wave packet and wavelength, not a resolved internal radius of the photon.

^aLecture timestamp: 00:36:49.

1.3 Exact transverse mechanics at large momentum

We now return to the relativistic difficulty. A rapidly rotating baton cannot be treated nonrelativistically merely by putting its center of mass at rest. Its ends still move relativistically relative to one another. The nonrelativistic approximation fails because of internal relative motion.

There is, however, a different limit. Choose a z -axis and boost the entire system to a very large positive longitudinal momentum P_z . Keep its transverse momentum

$$p_{\perp}^2 = p_x^2 + p_y^2 \tag{1.16}$$

finite. The transverse projection then admits an exactly nonrelativistic Hamiltonian form in the $P_z \rightarrow \infty$ limit, even if the unboosted internal motion was relativistic.

Question from the audience. Does infinite momentum mean that the massive system actually reaches the speed of light?^a

Response. No finite boost reaches it. One constructs quantities with well-defined limits as P_z becomes arbitrarily large. The phrase names a limiting procedure, not a finite physical state with $v = c$.

^aLecture timestamp: 00:40:43.

Question from the audience. Which axis must be chosen, and what if the object rotates about or across that axis?^a

Response. Any axis may be chosen and the whole system boosted along it. What remains as explicit mechanics is the projection onto the perpendicular plane. A rotating rod may look foreshortened or sweep through that plane, but its transverse projection has the nonrelativistic form.

^aLecture timestamp: 00:41:14.

Strings have an additional property that makes this representation unusually powerful. In light-front gauge their longitudinal internal coordinates are not independent oscillators. The constraints determine them from the transverse motion. Knowing the transverse string is therefore enough to reconstruct the physical degrees of freedom.

Question from the audience. Does the absence of an independent longitudinal coordinate mean that the string is merely trapped in a thin sandwich?^a

Response. Lorentz contraction does make its longitudinal extent small in the boosted frame, but the statement is stronger. Any residual longitudinal motion is constrained by the transverse motion; it cannot be specified independently.

^aLecture timestamp: 00:42:26.

Question from the audience. Is this reduction to transverse data related to the holographic principle?^a

Response. It is closely connected to the line of thought that led to holographic ideas, although pursuing that connection would take us away from the present calculation. Here it serves only to justify working with the transverse string.

^aLecture timestamp: 00:43:12.

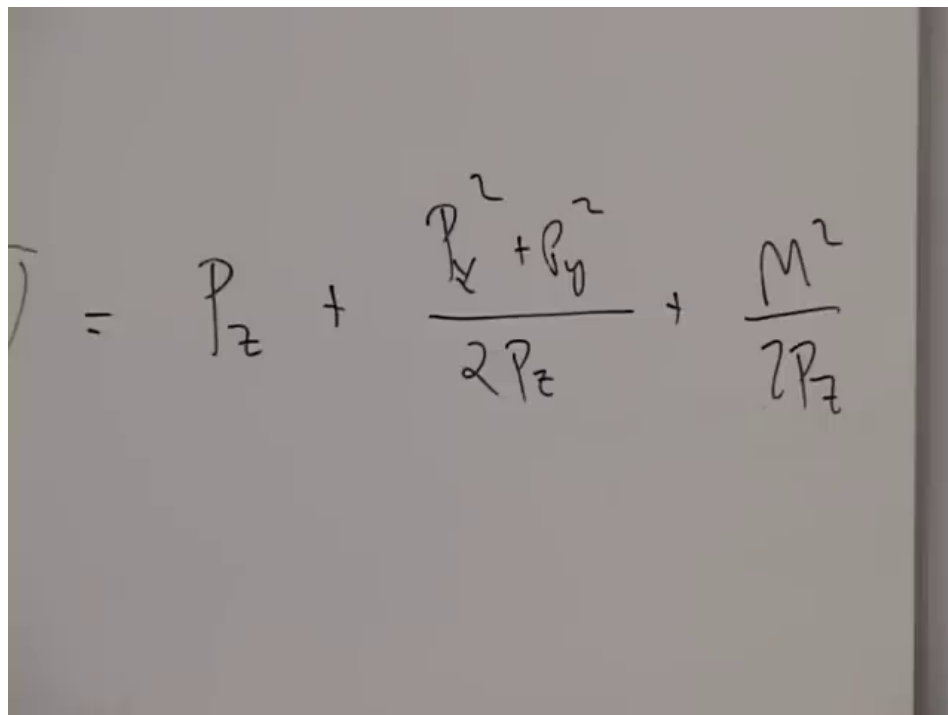
1.3.1 Subtracting the boost and rescaling time

Let M be the invariant mass of the complete system, including constituent masses, kinetic energy, and binding energy. In units $c = 1$,

$$E = \sqrt{P_z^2 + p_\perp^2 + M^2}. \quad (1.17)$$

For P_z much larger than $p_\perp$ and M ,

$$E = P_z + \frac{p_\perp^2 + M^2}{2P_z} + O(P_z^{-3}). \quad (1.18)$$



$$E = P_z + \frac{p_{\perp}^2 + M^2}{2P_z} + O(P_z^{-3})$$

Figure 1.8: The completed large- P_z expansion, with transverse momentum and invariant mass retained.

Lecture frame: 00:47:30.

The leading term is large, but it carries no information about the internal state. In a process that conserves P_z , such as a boosted atom changing between internal levels without changing its total longitudinal momentum, that term cancels from every energy difference. We therefore study

$$E - P_z = \frac{p_{\perp}^2 + M^2}{2P_z} + O(P_z^{-3}). \quad (1.19)$$

The denominator has a physical meaning. Since the Hamiltonian generates time evolution, the small residual energy says that internal evolution appears slow in laboratory time. This is time dilation. To retain the internal motion, introduce a rescaled evolution parameter τ , or equivalently multiply the residual Hamiltonian by P_z :

$$H_{\perp} \equiv \lim_{P_z \rightarrow \infty} P_z(E - P_z) = \frac{1}{2}p_{\perp}^2 + \frac{1}{2}M^2. \quad (1.20)$$

$$(E - P_z) = \frac{p_x^2 + p_y^2}{2P_z} + \frac{M^2}{2P_z}$$

$$H = i \frac{\partial}{\partial t}$$

 (a) $H = i\partial_t$ and the slowed internal evolution.

$$(E - P_z) P_z = \frac{p_x^2 + p_y^2}{2} + \frac{M^2}{2}$$

(b) The finite transverse Hamiltonian after time rescaling.

Figure 1.9: From the time-dilated residual energy to a finite two-dimensional Hamiltonian.

Lecture frame: 00:49:20 and 00:52:40.

This is precisely the form of a unit-mass particle moving in two dimensions: kinetic energy $p_{\perp}^2/2$ plus an internal energy $M^2/2$. The unusual feature is the square. In this description, the quantity playing the role of rest or binding energy is proportional to invariant mass squared.¹

Question from the audience. What happened to the longitudinal momenta of the individual parts of a composite system?^a

Response. For a generic composite object one must retain the fractions of total P_z carried by its constituents. The formula above uses the total center-of-mass momentum. String theory is special because its constraints determine the relative longitudinal motion from the transverse degrees of freedom, so no independent longitudinal oscillator has to be added.

^aLecture timestamp: 00:51:30.

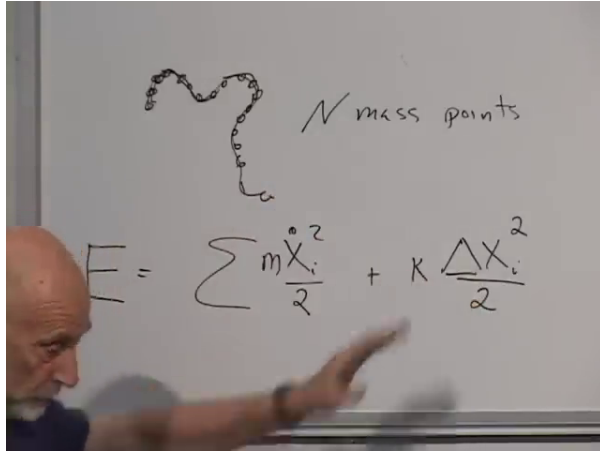
1.4 A discrete open string

The boosted string now moves explicitly only in the transverse plane. Label its two coordinates X and Y , and first approximate it by N mass points joined by $N - 1$ springs. If μ is the analog mass of each point and k the nearest-neighbor spring constant, the energy is

$$E_N = \sum_i \left[\frac{\mu}{2} (\dot{X}_i^2 + \dot{Y}_i^2) + \frac{k}{2} ((\Delta X_i)^2 + (\Delta Y_i)^2) \right]. \quad (1.21)$$

Every X -term has an identical Y -term because the transverse plane has two directions. The index i labels a point along the chain; it is not a spatial coordinate.

¹Editorial clarification: With conventional light-front variables $P^{\pm} = (E \pm P_z)/\sqrt{2}$, the exact mass-shell relation is $2P^+P^- = p_{\perp}^2 + M^2$. Equation (1.20) is the lecture's large- P_z normalization of the same invariant relation.



(a) The bead chain and its discrete energy.

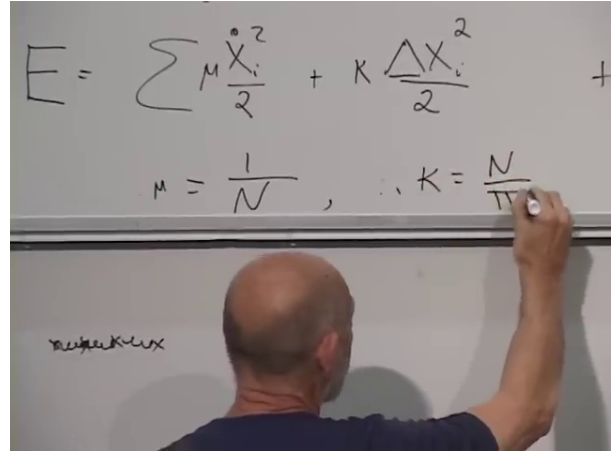

 (b) The large- N scaling of μ and k .

Figure 1.10: The regulated open string before taking the continuum limit.

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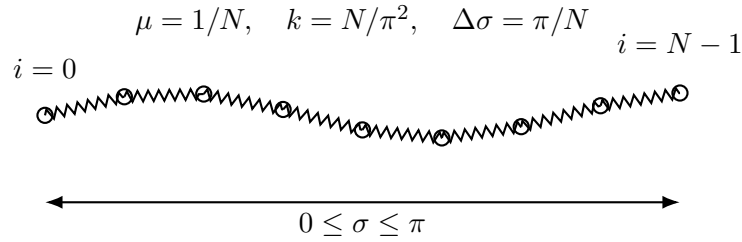


Figure 1.11: A clean regulator picture. The points become lighter while the microscopic springs become stiffer as the chain is refined.

The parameter σ labels position along the chain. Choosing $0 \leq \sigma \leq \pi$ is a convention, not a physical length measurement. It reminds us that the string has two endpoints. Later a closed string will be labeled periodically over $0 \leq \sigma < 2\pi$.

The continuum limit must keep the collective mechanics finite. Fixed bead mass would make the total analog mass diverge, so choose

$$\mu = \frac{1}{N}. \quad (1.22)$$

Then $N\mu = 1$. This is not the relativistic rest mass M ; it is the inertial coefficient in the transverse analog model.

Springs in series become softer as their number increases. To preserve a finite stiffness for the whole chain, each microscopic spring must become stiffer:

$$k = \frac{N}{\pi^2} = \frac{1}{\pi \Delta\sigma}. \quad (1.23)$$

Only $k \propto N$ is essential. The factor of π^2 is a normalization chosen to remove stray factors of π from the continuum formulas.

Question from the audience. Is μ the physical rest mass of each string point?^a

Response. No. It is the nonrelativistic analog mass appearing in the transverse chain. Calling it M would confuse it with the invariant rest mass of the entire relativistic string state. Its only present role is to produce the correct continuum inertia.

^aLecture timestamp: 01:10:34.

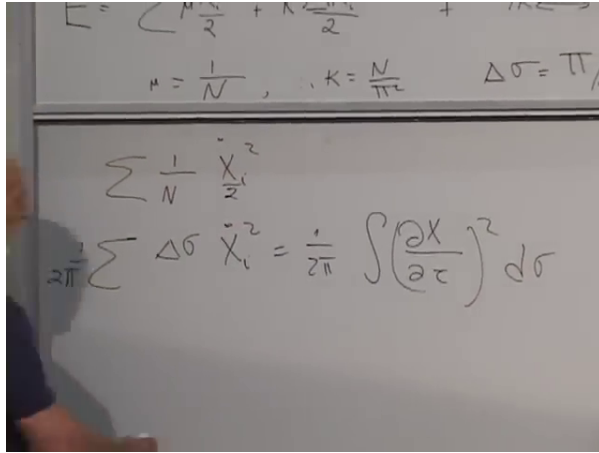
1.5 The continuum wave system

Apply the rules prepared at the beginning. Since $1/N = \Delta\sigma/\pi$,

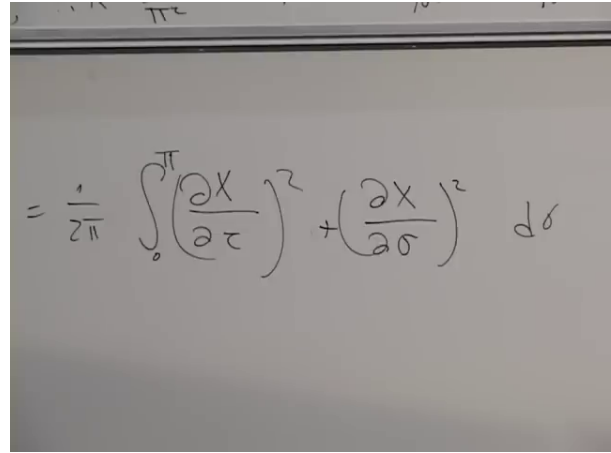
$$\sum_i \frac{\mu}{2} \dot{X}_i^2 = \frac{1}{2\pi} \sum_i \dot{X}_i^2 \Delta\sigma \longrightarrow \frac{1}{2\pi} \int_0^\pi (\partial_\tau X)^2 d\sigma, \quad (1.24)$$

$$\sum_i \frac{k}{2} (\Delta X_i)^2 \longrightarrow \frac{1}{2\pi} \int_0^\pi (\partial_\sigma X)^2 d\sigma. \quad (1.25)$$

For the second line, use $\Delta X_i = \partial_\sigma X \Delta\sigma$ together with $k = 1/(\pi\Delta\sigma)$. The same calculation applies to Y .



(a) Sum-to-integral and difference-to-derivative replacements.



(b) The completed continuum energy.

Figure 1.12: The discrete chain becoming a field on the σ -interval.

Lecture frame: 01:05:40 and 01:07:00.

The resulting energy and Lagrangian are

$$E = \frac{1}{2\pi} \int_0^\pi d\sigma \left[(\partial_\tau X)^2 + (\partial_\tau Y)^2 + (\partial_\sigma X)^2 + (\partial_\sigma Y)^2 \right], \quad (1.26)$$

$$L = \frac{1}{2\pi} \int_0^\pi d\sigma \left[(\partial_\tau X)^2 + (\partial_\tau Y)^2 - (\partial_\sigma X)^2 - (\partial_\sigma Y)^2 \right]. \quad (1.27)$$

The image shows a handwritten equation on a chalkboard. At the top left, there is a small expression $\frac{1}{N} \sum \frac{X_i^2}{2}$. Below it, a large equation is written: $\mathcal{L} = \frac{1}{2\pi} \int_0^\pi \left(\left(\frac{\partial X}{\partial \tau} \right)^2 - \left(\frac{\partial X}{\partial \sigma} \right)^2 \right) d\sigma$. To the right of the equation, there is a small sketch of a wave pulse on a horizontal line.

Figure 1.13: The continuum Lagrangian with kinetic and stretching terms carrying opposite signs.
Lecture frame: 01:08:48.

If we rename σ as an ordinary spatial coordinate and X as a field ϕ , these are the energy and Lagrangian of a one-dimensional wave field. Varying (1.27) gives

$$\left(\partial_\tau^2 - \partial_\sigma^2 \right) X = 0, \quad \left(\partial_\tau^2 - \partial_\sigma^2 \right) Y = 0. \quad (1.28)$$

Nothing in the immediate argument requires us to solve these equations directly. The useful picture is simply that waves travel left and right along the string and reflect at the endpoints. A Dirichlet reflection reverses the sign of the displacement; a Neumann reflection preserves it.

Question from the audience. What exactly is the time variable τ ?^a

Response. It is the rescaled time used to follow the internal motion in the large- P_z description. A clock boosted with the system runs slowly relative to laboratory time; τ removes that dilation so the internal oscillations retain a finite rate. For this transverse mechanics it can be treated simply as time.^b

^aLecture timestamp: 01:08:54.

^bEditorial clarification: The lecture informally calls τ proper time. For an extended string, the precise light-front evolution parameter is a gauge-fixed rescaled time rather than a single invariant proper time shared by every point. The intended time-dilation argument is unchanged.

When the center of mass is at rest in the transverse plane, the internal part of (1.26) is, in the normalization of (1.20), proportional to M^2 . That is why quantizing this apparently ordinary wave system will reveal the relativistic mass spectrum.

Question from the audience. Does $P_z(E - P_z)$ have a name?^a

Response. It is commonly called a light-cone or light-front energy. Light front better describes the null plane used as the evolution surface; the traditional terminology is not important for the calculation.

^aLecture timestamp: 01:11:00.

Question from the audience. Is the word cone referring to the usual spacetime light cone, and did Dirac introduce this terminology?^a

Response. The null-coordinate construction can be drawn with 45-degree axes, but the evolution surface here is a light front rather than the full expanding cone. Dirac developed related forms of relativistic dynamics; the historical naming question was left open and does not affect the equations.

^aLecture timestamp: 01:11:17.

1.6 Why a free endpoint is Neumann

We can now derive rather than postulate the endpoint condition. Newton's law is legitimate because the boosted transverse mechanics has exactly the nonrelativistic form. An interior mass point is pulled by springs on both sides. The final point is pulled only by the spring joining it to its neighbor.

Question from the audience. Why are we allowed to use Newton's law while discussing a relativistic string?^a

Response. We are not applying Newtonian mechanics to the full unboosted spacetime motion. We are applying it to the transverse variables in the infinite-momentum description, whose Hamiltonian has exactly the nonrelativistic form.

^aLecture timestamp: 01:14:52.

At the right endpoint, Hooke's law gives, up to orientation,

$$F_{\text{end}} = -k \Delta X = -\frac{1}{\pi \Delta \sigma} (\partial_\sigma X \Delta \sigma) = -\frac{1}{\pi} \partial_\sigma X. \quad (1.29)$$

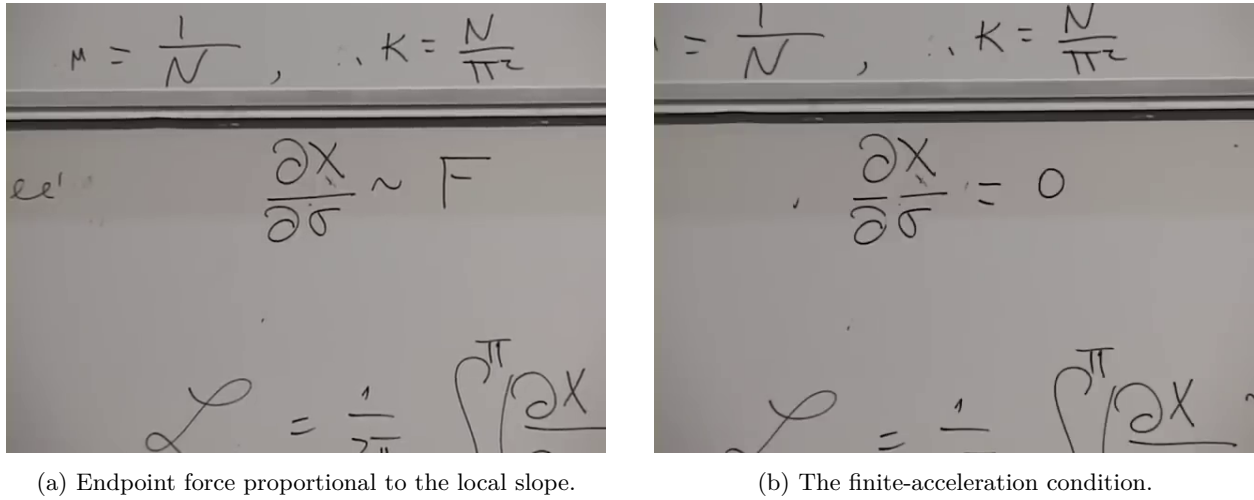


Figure 1.14: The Newton-law derivation of the free-end boundary condition.

Lecture frame: 01:18:20 and 01:20:10.

Newton's equation for the final point is

$$\mu \partial_\tau^2 X = F_{\text{end}}, \quad \mu = \frac{1}{N} = \frac{\Delta\sigma}{\pi}. \quad (1.30)$$

Combining (1.29) and (1.30),

$$\partial_\sigma X = -\Delta\sigma \partial_\tau^2 X \quad \text{at the right endpoint,} \quad (1.31)$$

with the force sign reversed at the left endpoint. If $\partial_\sigma X$ approached a nonzero constant while $\Delta\sigma \rightarrow 0$, the endpoint acceleration would diverge like N . A sensible continuum limit with finite acceleration therefore requires

$$\partial_\sigma X|_{0,\pi} = 0, \quad \partial_\sigma Y|_{0,\pi} = 0. \quad (1.32)$$

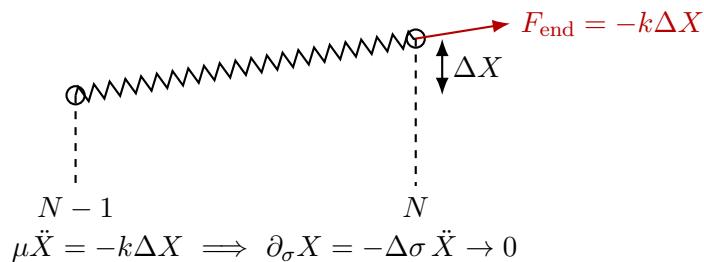


Figure 1.15: The last point has vanishing mass in the regulator limit. A finite acceleration is possible only when the endpoint slope vanishes.

The same argument at both ends and in both transverse directions produces Neumann boundary conditions for a free open string.

1.7 Normal modes of the open string

We now have a complete classical system: a Lagrangian and its boundary conditions. Quantization becomes simple after Fourier analysis. Because the endpoints are Neumann,

$$X(\sigma, \tau) = \sum_{n=0}^{\infty} X_n(\tau) \cos(n\sigma), \quad (1.33)$$

$$Y(\sigma, \tau) = \sum_{n=0}^{\infty} Y_n(\tau) \cos(n\sigma). \quad (1.34)$$

The cosines contain all σ -dependence. The coefficients contain all time-dependence; the string wiggles because $X_n(\tau)$ and $Y_n(\tau)$ change.

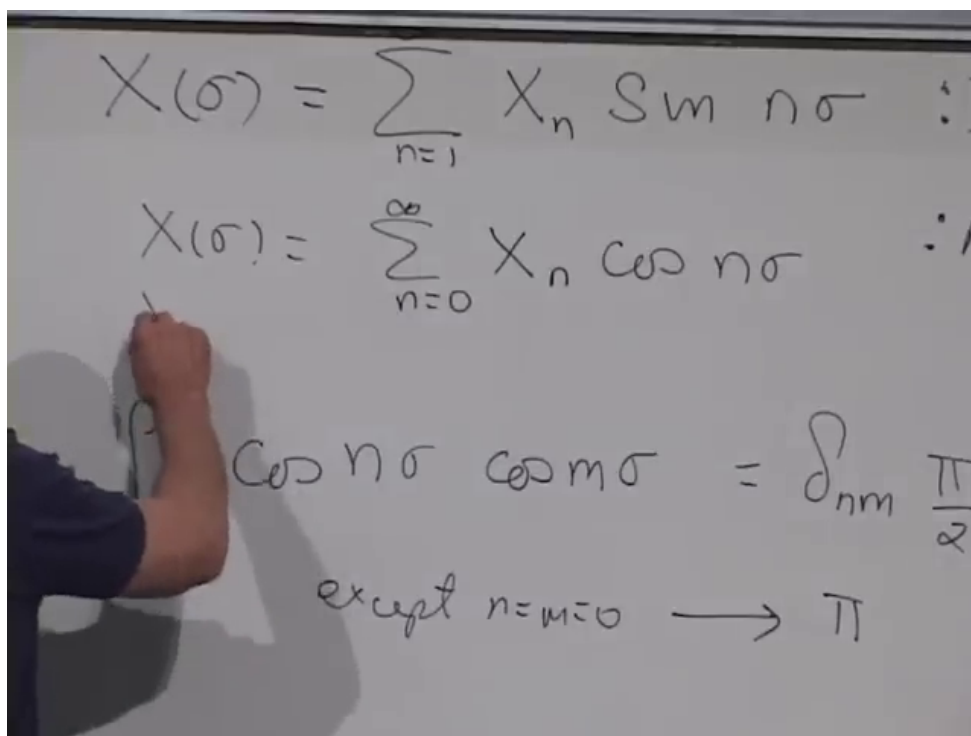


Figure 1.16: The Neumann mode expansion placed beside the orthogonality rule that will collapse the double sums.

Lecture frame: 01:21:45.

1.7.1 Kinetic term and the zero mode

Time differentiation gives

$$\partial_\tau X = \sum_{n=0}^{\infty} \dot{X}_n(\tau) \cos(n\sigma). \quad (1.35)$$

Squaring produces a double sum over n and m , but (1.10) removes every term with $n \neq m$. Treating the constant mode separately,

$$\frac{1}{2\pi} \int_0^\pi (\partial_\tau X)^2 d\sigma = \frac{1}{2} \dot{X}_0^2 + \frac{1}{4} \sum_{n=1}^\infty \dot{X}_n^2. \quad (1.36)$$

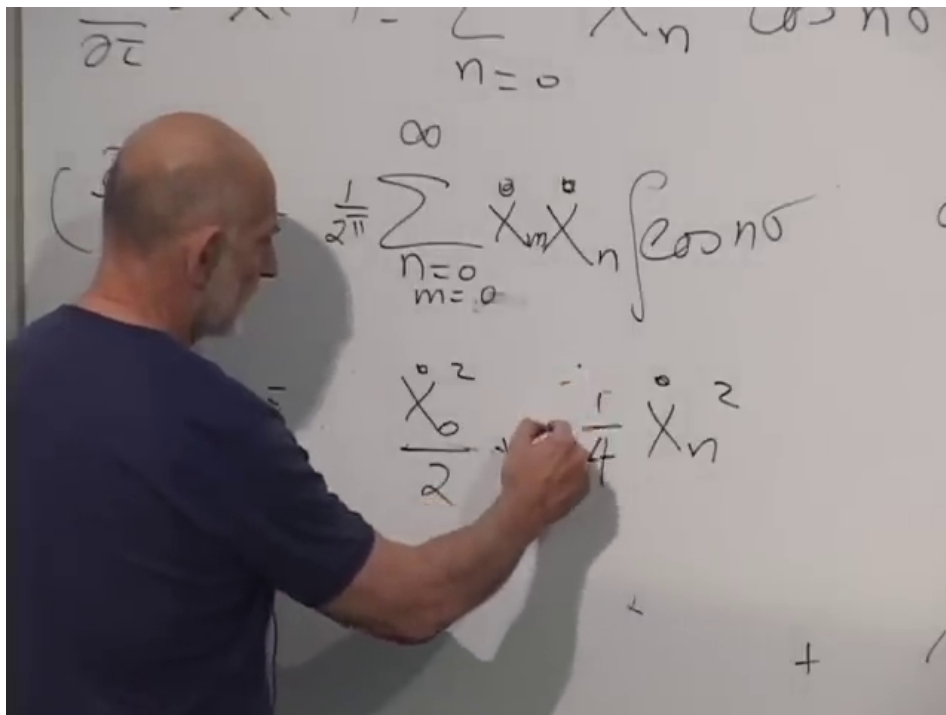


Figure 1.17: The orthogonality integral reducing the kinetic double sum to the zero mode plus a diagonal sum.

Lecture frame: 01:29:10.

The constant coefficient is the average position,

$$X_0(\tau) = \frac{1}{\pi} \int_0^\pi X(\sigma, \tau) d\sigma, \quad (1.37)$$

and similarly for Y_0 . With uniform analog mass density, these are the center-of-mass coordinates. They have kinetic energy but no restoring force: the entire string is free to move across the transverse plane.

1.7.2 Stretching term and oscillator frequencies

Spatial differentiation gives

$$\partial_\sigma X = - \sum_{n=1}^\infty n X_n(\tau) \sin(n\sigma). \quad (1.38)$$

The zero mode disappears because a uniform translation does not stretch the string. Using sine orthogonality,

$$\frac{1}{2\pi} \int_0^\pi (\partial_\sigma X)^2 d\sigma = \frac{1}{4} \sum_{n=1}^\infty n^2 X_n^2. \quad (1.39)$$

$$+ \frac{1}{2\pi} \sum_{nm} n m X_n X_m \int_0^\pi \sin n\sigma \sin m\sigma d\sigma$$

$$+ (x \rightarrow y)$$

$$\int_0^\pi \sin n\sigma \sin m\sigma d\sigma = \frac{\pi}{2} \delta_{nm}$$

Figure 1.18: Differentiating the cosine series produces the sine double sum; orthogonality again enforces $n = m$.

Lecture frame: 01:31:40.

The X -sector Lagrangian is therefore

$$L_X = \frac{1}{2} \dot{X}_0^2 + \frac{1}{4} \sum_{n=1}^\infty (\dot{X}_n^2 - n^2 X_n^2), \quad (1.40)$$

with an identical expression for Y . The corresponding energy has a plus sign in front of $n^2 X_n^2$. Every nonzero mode is an independent harmonic oscillator, and comparison with (1.14) gives

$$\omega_n = n, \quad n = 1, 2, 3, \dots \quad (1.41)$$

Handwritten notes on a chalkboard:

- At the top left, $n=0$ is written.
- Below it, $\omega_n = n$ is written.
- In the center, a large curly bracket groups the following expression:

$$\frac{\dot{X}_0^2}{2} + \left(\sum_n \frac{1}{4} \dot{X}_n^2 - \frac{1}{4} \sum_n n^2 X_n^2 \right)$$
- Below the bracketed expression, there is a plus sign, a slash, and an arrow pointing to a 'y' in parentheses: $+ / \rightarrow y)$.
- At the bottom right, the expression $\omega = \pi/N$ is written, followed by $\frac{1}{N} =$.

Figure 1.19: The decoupled oscillator Lagrangian and the conclusion $\omega_n = n$, with the identical Y -tower indicated below.

Lecture frame: 01:35:50.

The normal modes are simply the harmonics of the string. The first bends slowly, the second has twice the frequency, the third three times, and so on. Orthogonality is what prevents one harmonic from coupling to another in the free theory.

Question from the audience. What happened to the $n = 0$ oscillator?^a

Response. It has zero frequency because it is not an internal oscillator at all. It translates the center of mass. No spring restores the whole string to a preferred transverse position, so the zero mode is a free particle coordinate.

^aLecture timestamp: 01:34:49.

Dropping the center-of-mass kinetic term leaves the internal energy

$$E_{\text{int}} = \frac{1}{4} \sum_{n=1}^{\infty} \left[\dot{X}_n^2 + n^2 X_n^2 + \dot{Y}_n^2 + n^2 Y_n^2 \right]. \quad (1.42)$$

In the light-front normalization this is proportional to the invariant mass squared of the string state. We have thus converted the relativistic mass problem into the quantum mechanics of two countable oscillator towers.

1.8 Quantized levels and particle behavior

Let $a_n^\dagger$ create one quantum in the X -oscillator of frequency n , and $b_n^\dagger$ do the same for Y . The lecture postpones a systematic operator treatment, but this notation makes the level counting transparent. Relative to the ground-state offset,

$$\Delta E_{\text{int}} = \sum_{n=1}^{\infty} n (Q_n^X + Q_n^Y), \quad (1.43)$$

$$Q_n^X, Q_n^Y \in \{0, 1, 2, \dots\}.$$

where $\hbar = 1$. The integer n labels which oscillator is excited; Q_n counts how many quanta occupy it.²

The oscillator vacuum $|0\rangle$ is not empty spacetime. It is one open string with no internal oscillator excited. It has energy, momentum, and a definite place in the string spectrum.

At level one there are two states,

$$a_1^\dagger |0\rangle, \quad b_1^\dagger |0\rangle, \quad (1.44)$$

corresponding to the lowest harmonic polarized along X or Y . Linear combinations describe any transverse polarization.

At level two there are five states:

$$(a_1^\dagger)^2 |0\rangle, \quad a_1^\dagger b_1^\dagger |0\rangle, \quad (b_1^\dagger)^2 |0\rangle, \quad a_2^\dagger |0\rangle, \quad b_2^\dagger |0\rangle. \quad (1.45)$$

Three use two quanta of the first harmonic; two use one quantum of the second harmonic. Both constructions cost two units.

²Editorial clarification: Each oscillator also contributes a zero-point energy. The infinite sum requires regularization and supplies the intercept of the string mass formula. The lecture deliberately leaves the ground-state offset unspecified here and counts only excitation levels.

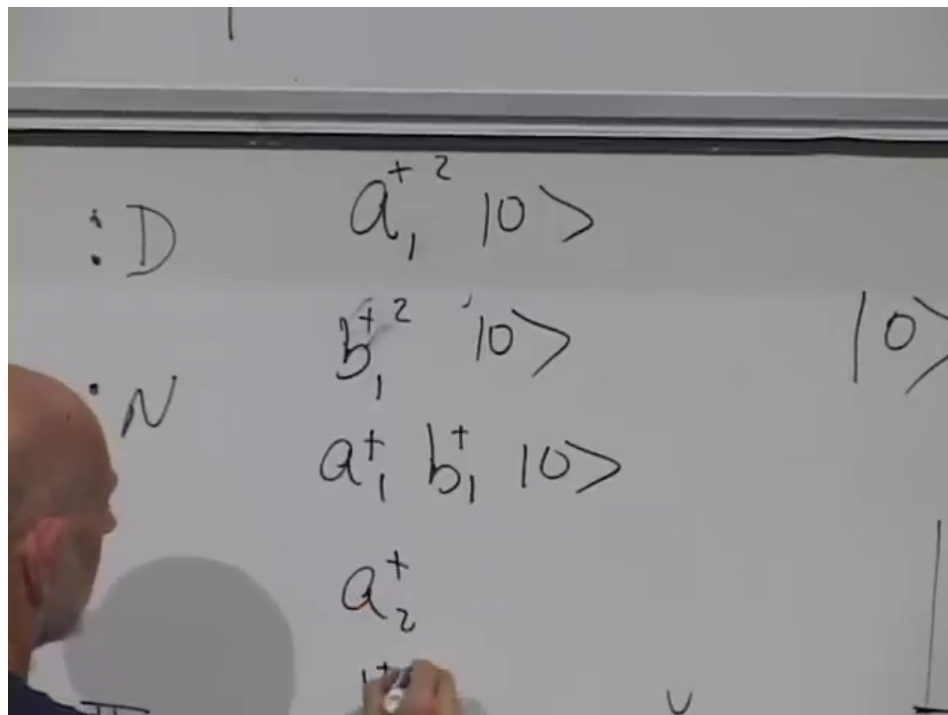


Figure 1.20: The five level-two states being recounted after an audience challenge: two double excitations, one mixed excitation, and two second-harmonic excitations.

Lecture frame: 01:44:00.

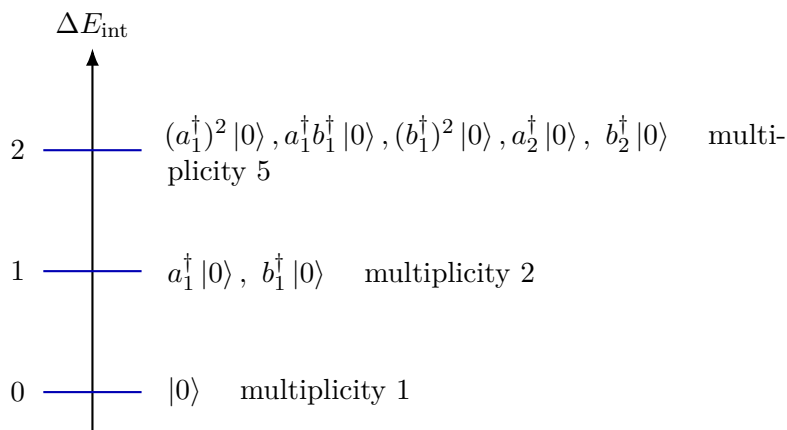


Figure 1.21: The first three excitation levels of the two-transverse-direction open string.

Question from the audience. Are there four or five states at the second excited level?^a

Response. Five. Exciting X_1 twice, exciting Y_1 twice, and exciting each once give three states. Exciting X_2 once or Y_2 once gives two more. The proposed exercise is to count the next two levels.

^aLecture timestamp: 01:43:04.

The essential conclusion is already visible. Successive levels are separated by finite gaps, and each has a finite multiplicity. The quantized string is therefore particle-like by the spectral criterion that

distinguished it from coffee, gas, or an ordinary violin string. Later we shall determine which states are physically welcome and which pathologies must be removed.

1.9 Why open-string interactions produce closed strings

An open string has two endpoints. Its elementary interactions join and split strings. In the hadronic picture one may imagine a quark at one end and an antiquark at the other; their annihilation joins two string segments. The fundamental strings of interest later need not carry that literal hadronic endpoint interpretation, but the local joining rule remains.

Now let one open string fluctuate until its own endpoints approach one another. Locally, an endpoint cannot know that the other endpoint belongs to the same string rather than a different one. If endpoints may join at all, they may join here, producing a closed loop.

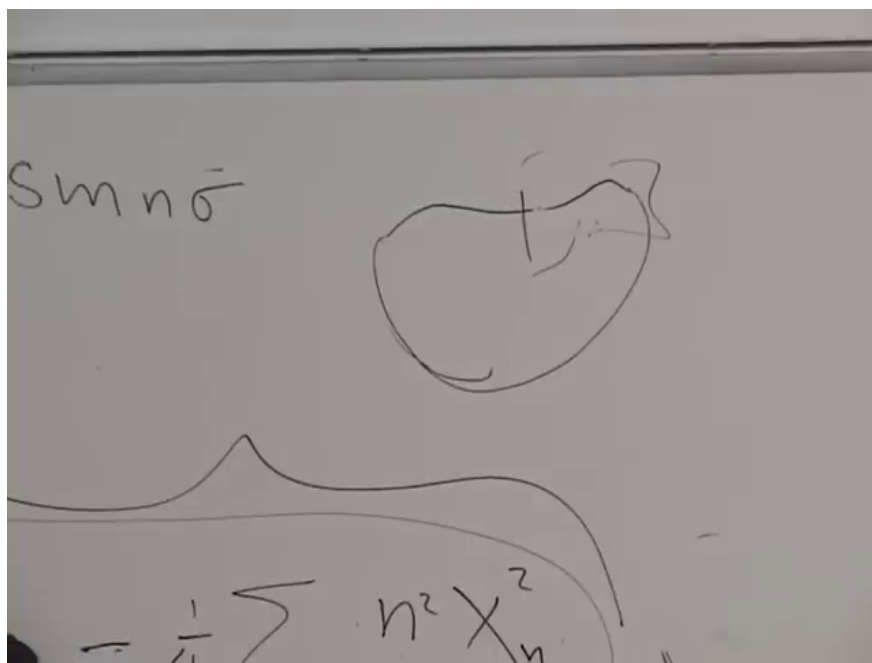


Figure 1.22: The board sketch of an open string fluctuating until its two endpoints meet and form a closed loop.

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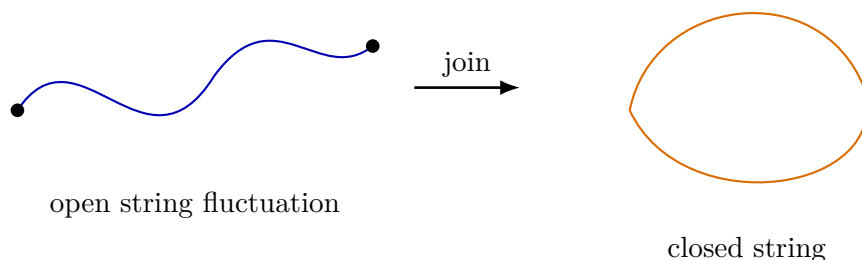


Figure 1.23: Once endpoint joining is allowed, an open string can close on itself.

Thus a theory may contain closed strings without open strings, but an interacting theory with open strings cannot consistently avoid closed strings. This asymmetry is the important prediction. The spectral identifications will be developed later, but the destination is already visible: massless open-string states often supply gauge particles such as photons, while the closed-string spectrum contains a massless spin-two state, the graviton. Closed strings, and therefore gravity, are not an optional decoration once string theory is interacting.